```
import pandas as pd
import numpy as np
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
uploaded = files.upload()
df = pd.read_csv(next(iter(uploaded)))
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving air_quality_dataset.csv to air_quality_dataset (3).csv
```

```
conditions = [
      df["PM2.5"] <= 35,
      (df["PM2.5"] > 35) & (df["PM2.5"] <= 75),
       df["PM2.5"] > 75
 ]
```

```
df["air_quality_dataset"] = np.select(
      conditions,
          ["Good", "Moderate", "Poor"],
              default="Unknown"
              )

X = df.drop("air_quality_dataset", axis=1)
y = df["air_quality_dataset"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
      X, y, test_size=0.2, random_state=0, stratify=y
      )
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465: Convergen
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

▾ LogisticRegression

```
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
```

```
y_pred = model.predict(X_test)
```

```
print("Model Trained Successfully!")
print("Test Accuracy:", accuracy_score(y_test, y_pred) * 100, "%")
```

```
Model Trained Successfully!
Test Accuracy: 93.0 %
```

```
new_sample = pd.DataFrame(
      [[40, 60, 90, 45, 12, 1.8, 0.8, 420, 2.1, 30]],
          columns=X.columns
          )
```

```
prediction = model.predict(new_sample)

print("Predicted Air Quality:", prediction[0])
```

```
Predicted Air Quality: Moderate
```